
CLINTON ENWEREM

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: [@coenwerem](https://github.com/coenwerem) | [LinkedIn](#) | [Scholar](#)

SUMMARY

Robotics Research & Software Engineer, Ph.D. (ECE, University of Maryland, 2026). Specialties in dexterous manipulation, risk-aware and safety-critical control, belief-space planning, reinforcement and imitation learning, and real-time QP control. **5 years** building manipulation software that runs on real multifingered hardware. 7 peer-reviewed and accepted papers. Seeking research scientist, research engineer, or applied ML roles.

SKILLS

Robotics: ROS/ROS2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, MoveIt2; **Control & Safe Learning:** Control Barrier Functions, Safety Filters, Safe & Constrained Reinforcement Learning, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Sim-to-Real Transfer; **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi; **Dev Tools:** git, Docker; **Robots & EoATs:** Fairino FR3, xArm7, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

WORK EXPERIENCE

Graduate Research Assistant

Institute for Systems Research (ISR), University of Maryland

Aug. 2021 – Present
College Park, MD

- Build and maintain the lab's Python and C++ grasp planning and closed-loop control stack. Deploy it on Fairino FR3 and xArm7 arms with LEAP Hand v1 and RealHand L6 end-effectors.
- Build expert data collection pipelines over synchronized tactile, proprioceptive, and visual signals. Train MLP, LSTM, and DP3 behavior cloning policies for grasp stabilization under partial tactile observability.

Robotics Trainee

Robotics & Artificial Intelligence Nigeria

Jan. 2020 – Feb. 2021
Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

PROJECTS

GDF, Planner-Free Certified Grasping (Project Page) | *Python, MuJoCo, CasADi*

2026

- Built a planner-free grasp execution controller from a softmin distance field over certified pregrasp configurations, filtered by a CBF-CLF quadratic program. Proved forward invariance of the safe set. Lifted **46 of 50** objects, retaining a median **94%** of the synthesized quality margin.
- Ported the controller to a **Unitree G1 humanoid** and a fixed-base arm by swapping only the robot description. QP solves in **0.09 ms** against a 20 ms control step.

EquiDexFlow, SE(3)-Equivariant Grasp Generation (Project Page, Code) | *Python, PyTorch*

Jun. 2026

- Trained an SE(3)-equivariant flow-matching grasp generator on 8,100 certified grasps across 81 objects, with friction-cone compliance holding by construction. **Zero** friction violations. Retargeted to a physical LEAP Hand by IK alone.

VNB-Grasp, Variational Neural Belief Parameterizations (Code) | *Python, PyTorch, MuJoCo*

Apr. 2026

- Designed a differentiable Gaussian-mixture contact belief with pathwise CVaR gradients, and released a MuJoCo benchmark that scores friction sensitivity and perturbation tolerance. Cut belief propagation **over 10x**.

URDF Kinematics Toolkit (Code) | *Python*

Oct. 2025

- Built a URDF parsing and kinematics library with forward and inverse kinematics, damped least-squares Jacobian resolution, condition-number singularity margins, and kinematics-aware configuration-space sampling.

SELECTED PUBLICATIONS

- **C. Enwerem**, J. S. Baras, C. Belta. "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Manuscript*, 2026. Project page.
- **C. Enwerem**, J. S. Baras, C. Belta. "FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis." *Preprint*, 2026. [arXiv:2607.25049](https://arxiv.org/abs/2607.25049).
- Also to appear 2026 at IROS and CDC, plus 5 peer-reviewed first-author papers (CoDIT'23, L-CSS'24, ECC'24, CDC'24, CDC'25).

EDUCATION

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018